

Three Roads: Choose Your Stack with AI

Written Rationale

I chose Astro HTML CSS and minimal JS.

Here's why: I did not want something that would be too complex for the current portfolio iteration but I also did not want something too simple that would not allow expansion headroom for future iteration. This stack works well because it accounts for a current characteristic of the content. I have multiple case studies that share same structure (heading and layout). I chose a stack that will allow me to have a component that can be repeated instead of manually repeating code across each case study which would be difficult to maintain. I chose HTML and CSS since I already know these technologies and they allow me to code with AI and achieve my portfolio look and feel without introducing a complex framework like React/Tailwind. Minimal JS is included as the tool to allow for interaction where necessary like expandable sections but it is only used where needed.

I did not choose a no code solution because I did not want something too simple I wanted room for customization and control over the look and feel. No code is fast but that is paid for in less control over implementation. I also did not choose a framework because the current solution is static with links and pictures and a gif. This does not need a complex implementation stack. I chose Astro for the components so that maintainability and scaling would be more feasible.

Can I maintain this? Yes, I am hosting the app on Netlify and the set-up of the website is straight forward. There is no backend which makes things simpler and coding of the initial site outside of Astro is trivial save for the design considerations.

Does it show my work well? Yes, it allows me to implement the current portfolio vision. CSS and JS will allow for custom styling customization to achieve the exact look and feel needed. The work is allowed to be the focus while looking professional and clean.

Stated explicitly I do not need a backend for my portfolio.

Considered Options and Trade-offs

	HTML/CSS/JS	Astro	Astro + React + Tailwind
Simplicity	★★★★★	★★★★★	★★
Visual control	★★★★★	★★★★★	★★★★★
Your current portfolio	★★★★★	★★★★★	★★★★★
Reusable case structure	★★	★★★★★	★★★★★
Future interactive work	★★★	★★★★★½	★★★★★
Learning overhead	★	★★½	★★★★
Maintenance	★★★★★	★★★★★	★★★
Risk of overbuilding	★	★½	★★★★
Headroom	★★★	★★★★★½	★★★★★

	No-code	Code with AI	Framework
Build speed	★★★★★	★★★★★	★★
Visual control	★★★★★	★★★★★	★★★★★
Learning value	★	★★★★★	★★★★★
Easy to maintain	★★★★★	★★★★★	★★★
Easy to explain	★★★★★	★★★★★	★★★
Free long-term	Depends on platform	★★★★★	★★★★★
Fits your evidence	★★★★★	★★★★★	★★★★★
Risk of overbuilding	Low	Low	High